

Advanced Data Structures and Algorithm Analysis

SITAMS CSE (AI & ML) • R23 • II Year • III Semester • Course Code: 23CSE231T

Source basis: Official SITAMS R23 CSE (AI & ML) syllabus uploaded for this Study Hub. This concise revision guide preserves the syllabus topic statements and adds a study checklist; it is not a replacement for prescribed textbooks or faculty notes.

Unit I: INTRODUCTION

Syllabus topics

- Introduction to Algorithm Analysis, Space and Time Complexity analysis, Asymptotic Notations
- AVL Trees – Creation, Insertion, Deletion operations and Applications
- B-Trees – Creation, Insertion, Deletion operations and Applications

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

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Unit 2: HEAP TREES & DIVIDE AND CONQUER

Syllabus topics

- Heap Trees (Priority Queues) – Min and Max Heaps, Operations and Applications Graphs – Terminology, Representations, Basic Search and Traversals, Connected Components and Biconnected Components, applications
- Divide and Conquer: The General Method, Quick Sort, Merge Sort, Strassen's matrix multiplication, Convex Hull

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Unit 3: GREEDY METHOD & DYNAMIC PROGRAMMING

Syllabus topics

- Greedy Method: General Method, Job Sequencing with deadlines, Knapsack Problem, Minimum cost spanning trees, Single Source Shortest Paths
- Dynamic Programming: General Method, All pairs shortest paths, Single Source Shortest Paths – General Weights (Bellman Ford Algorithm), Optimal Binary Search Trees, 0/1 Knapsack, String Editing, Travelling Salesperson problem

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Unit 4: BACKTRACKING & BRANCH AND BOUND

Syllabus topics

- Backtracking: General Method, 8-Queens Problem, Sum of Subsets problem, Graph Coloring, 0/1 Knapsack Problem
- Branch and Bound: The General Method, 0/1 Knapsack Problem, Travelling Salesperson problem

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Unit 5: NP HARD, NP HARD GRAPH PROBLEMS & NP HARD SCHEDULING PROBLEMS

Syllabus topics

- NP Hard and NP Complete Problems: Basic Concepts, Cook's theorem NP Hard Graph Problems: Clique Decision Problem (CDP), Chromatic Number Decision Problem (CNDP), Traveling Salesperson Decision Problem (TSP) NP Hard Scheduling Problems: Scheduling Identical Processors, Job Shop Scheduling

Revision checklist

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Study Hub status

This guide is syllabus-derived. External resources and quiz questions are tracked separately and are only labelled verified after source checking.